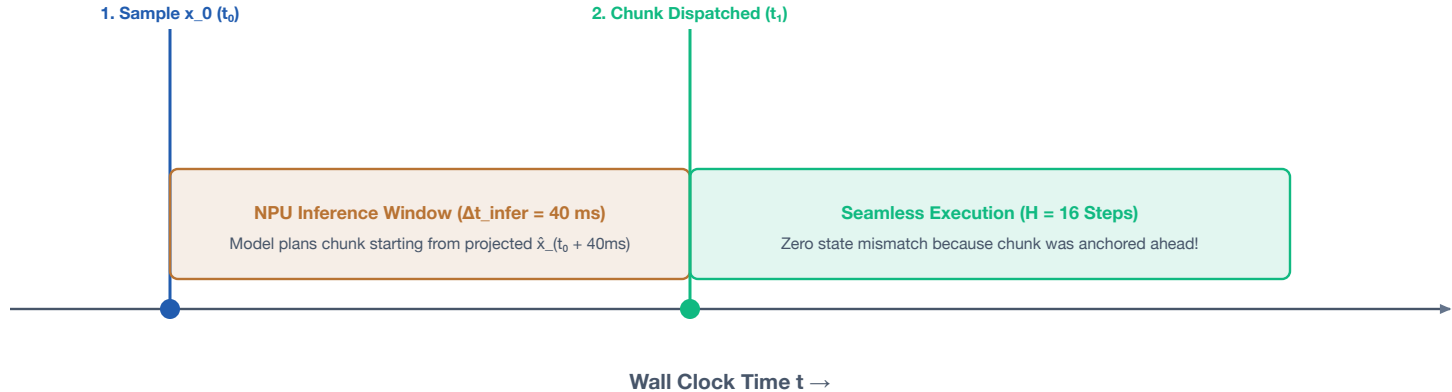


ASYNCHRONOUS STATE RE-ANCHORING HANDSHAKE

Planning Ahead: Projecting State $\hat{x}_t(t + \Delta t_{\text{infer}})$ During Model Inference to Eliminate Lag Shock



THE RE-ANCHORING INVARIANT

1. Naïve planners anchor at $x(t_0)$; by the time inference completes at t_1 , the robot has moved to $x(t_1)$, causing a violent positional snap.
2. Defensive Physical AI architectures project the robot forward using the real-time MCU state estimator: $\hat{x}(t_0 + \Delta t_{\text{infer}})$.
3. The chunk arrives exactly when the robot enters the projected state, ensuring C^2 continuity.